

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTITC403 - DISCRETE MATHEMATICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

1 Let $A = \{2, \{4, 5\}, 8\}$. Which statement is correct?

1. 5 is an element of A.
2. $\{5\}$ is an element of A.
3. $\{4, 5\}$ is an element of A.
4. $\{5\}$ is a subset of A.

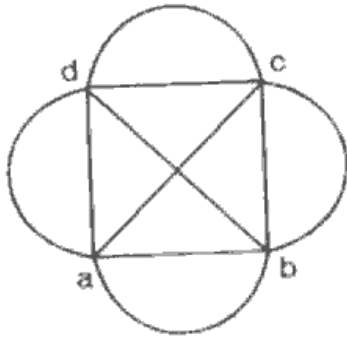
2 Select correct statement as an equivalent statement that does not use the if . . . then connective. Remember that $p \rightarrow q$ is equivalent to $\sim p \vee q$. If the sun comes out tomorrow, the roses will open.

1. The sun does not come out tomorrow and the roses will not open.
2. The sun does not come out tomorrow or the roses will not open.
3. The sun does not come out tomorrow or the roses will open.
4. The sun comes out tomorrow and the roses will not open.

3 The proposition $p \wedge (\sim p \vee q)$ is a.

1. a tautology
2. a contradiction
3. logically equivalent to $p \wedge q$
4. logically equivalent to $p \vee q$

- 4 Does the following graph has Euler circuit or path



1. Abcd is Euler circuit and abcd is Euler path
 2. Does not have Euler circuit but has path
 3. Does not have Euler path but has Euler circuit
 4. Does not exist Euler path as well as Euler circuit
- 5 Consider $X = \{2, 3, 6\}$; $Y = \{8, 9\}$ and $R = \{(2, 8), (2, 9), (6, 9), (3, 9)\}$. The complement of relation is
1. $\{(2, 8), (3, 8), (6, 9), (6, 9)\}$
 2. $\{(2, 9), (6, 8), (3, 8)\}$
 3. $\{(2, 9), (6, 8), (3, 8)\}$
 4. $\{(3, 8), (6, 8)\}$
- 6 Consider $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (1, 4), (2, 3), (2, 4), (3, 4)\}$. The domain of R is
1. $\{1, 2, 3\}$
 2. $\{1, 2, 3, 4\}$
 3. $\{2, 3, 4\}$
 4. $\{1, 2\}$
- 7 If $A = \{1, 2, 3, 4, 6\} = B$ and aRb iff $a|b$ then relation R is _____
1. $\{(1, 2), (2, 4), (3, 6)\}$
 2. $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 6), (2, 4), (2, 6), (3, 6)\}$
 3. $\{(1, 1), (2, 2), (2, 4), (2, 6), (3, 6)\}$
 4. $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (6, 6)\}$
- 8 Find the range for $f(x) = -x^2 + 3$, given the domain $\{-3, 0, 4\}$
1. $\{12, -3, 19\}$
 2. $\{-6, 3, -13\}$
 3. $\{9, -3, -5\}$
 4. $\{-12, -3, 19\}$
- 9 A binary tree in which every level, except possibly the last, is completely filled and all nodes are as far left as possible is called:
1. Full binary tree
 2. Perfect binary tree
 3. Complete binary tree
 4. Balanced binary tree

10 **A group is a set G equipped with a binary operation $*$ such that:**

1. $*$ is associative
2. There is an identity element in G
3. Every element has an inverse in G
4. All of the above

11 **If every element of a group is its own inverse, then the group is:**

1. Cyclic
2. Abelian
3. Finite
4. Trivial

12 **From a set of 8 different books, how many ways can 3 books be selected and arranged on a shelf?**

1. 336
2. 56
3. 512
4. 24

Q2. Solve the following.

A) In the class of 55 students the number of students studying different subjects are as given below: Maths 23, Physics 24, Chemistry 19, Maths + Physics 12, Maths + Chemistry 9, Physics + Chemistry 7, all three subjects 4. Find the number of students who have taken

1. At least one subject.
2. Exactly one subject.

B) Use De Morgan's laws to find the negation of each of the following statements.

1. Kwame will take a job in industry or go to graduate school.
2. Yoshiko knows Java and calculus.
3. James is young and strong.
4. Rita will move to Oregon or Washington.

Q3. Solve the following.

A) Prove the mathematical induction for $n \geq 1$
 $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$

B) Let $A = \{0, 1, 2, 3\}$ Decide whether or not the following are function from A to A . Draw the diagram for the function. If not write the reason.

1. $f = \{(0, 1), (1, 2), (2, 3), (3, 0)\}$
2. $f = \{(1, 0), (2, 1), (3, 2), (0, 3)\}$
3. $f = \{(0, 1), (1, 0), (2, 1), (3, 0)\}$
4. $f = \{(1, 2), (2, 3)\}$

Q4. Solve Any Two of the following.

A) Solve the recurrence relation $F_n = 5F_{n-1} - 6F_{n-2}$ where $F_0 = 1$ and $F_1 = 4$

B) Suppose repetitions are permitted: 6

1. How many ways three-digit number can be formed from six digits 2, 3, 4, 5, 7 and 9?
2. How many of these numbers are less than 400 ?

C) Find the number of arrangements that can be made out of the letters: 6

1. ASSASSINATION
2. GANESHPURI
3. BANANA

Q5. Solve Any Two of the following.

A) Let $A = \{1, 2, 3\}$ and consider two reflexive relations $R = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 3)\}$ and $S = \{(1, 1), (1, 2), (2, 2), (3, 2), (3, 3)\}$. Determine the following relations are reflexive or irreflexive? 6

1. R^{-1}
2. R^c

3. $R \cap S$

4. $R \cup S$

5. $R \circ S$

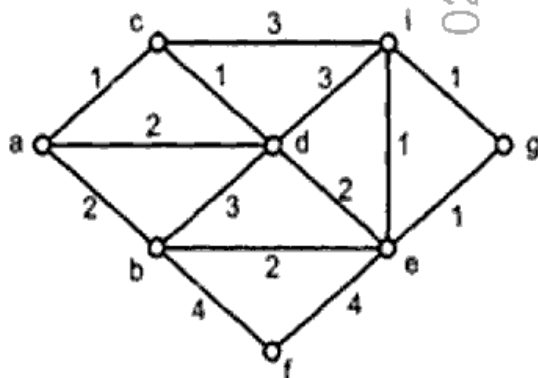
B) Define with examples 6

1. Monoid
2. Group
3. Abelian Group
4. Ring

C) Let R be the relation on the set $A = \{5, 6, 8, 10, 28, 36, 48\}$. Let $R = \{(a, b) \mid a \text{ is a divisor of } b\}$. Draw the Hasse diagram compare with diagram. Determine whether R is equivalent relation. <https://www.batuonline.com> 6

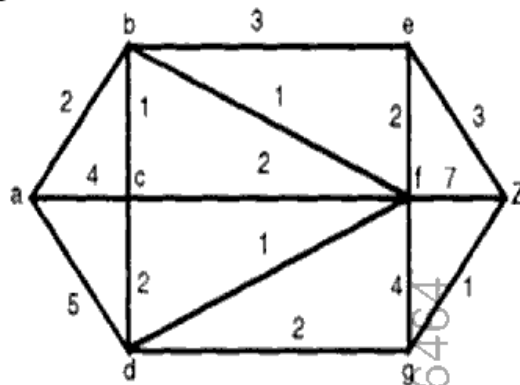
Q6. Solve Any Two of the following.

A) Use Prim's algorithm for finding minimum spanning tree in fig starting at node a. 6



- B) Find the shortest path between a — z for the given graph using Dijkstra's algorithm:

6



Define with example

1. Regular Graph
2. Complete Graph
3. Spanning Trees
4. Bipartite Graph

6

*** End ***